

AI Resume-Ready Projects Roadmap

Step-by-step build guides for 5 production-grade AI projects to showcase in 2025.

1. Multi-Agent Data Ops for Content Teams

- Goal: Turn a PRD (product requirements doc) into a polished blog post using 4 chained agents.
- Agents: Researcher (sources) → Writer (draft) → Fact-checker (verify) → Style-polisher (tone).
- Setup: Next.js + Supabase + LangGraph/CrewAI, OpenAI/Anthropic models.
- Ops Features: Logging in Supabase, rubric scoring, rollback if fact-check fails.
- Demo: Visual timeline PRD → Draft → Fact-check → Polished + metrics dashboard.

2. Drift-Aware Retraining Pipeline

- Goal: Keep AI system fresh as data/behavior shifts.
- Detects: Embedding drift, refusal/toxicity spikes, accuracy drops.
- Triggers: Re-index docs, refresh retrieval, fine-tune adapter.
- Setup: Python + pgvector/Supabase, Prometheus/Grafana for metrics.
- Demo: Chart showing accuracy↑ and cost↓ after refresh.

3. Model Router with Budgets

- Goal: Route easy queries to cheap models, complex ones to smart models.
- Policy: 80% → cheap, 15% → mid, 5% → expert models.
- Setup: Next.js API gateway, GPT-3.5/Claude Haiku/GPT-4.1 routing.
- Ops Features: Difficulty estimator, monthly budget ceiling, alerts.
- Demo: Heatmap of cost vs quality per route.

4. Guardrail & Red-Team Harness

- Goal: Stress-test AI feature with adversarial prompts and monitor guardrail performance.
- Setup: Python/Node + Supabase/SQL, moderation API, regex scrubbers.
- Ops: Store adversarial test cases (JSON), replay nightly.
- Metrics: Jailbreak success rate, moderation FP/FN, alerts.
- Demo: Dashboard showing red-team attempts vs blocked + incident log.

5. Evaluation-as-a-Service (EaaS)

- Goal: Platform for uploading prompt/output sets with rubrics, runs automated evals.
- Setup: Next.js frontend, Supabase backend, Python scoring worker.
- Features: Upload CSV, run evals, store results with experiment ID.
- Ops: CI/CD hook (block deploy if below threshold), versioned trends.
- Demo: Dashboard with eval runs, trend comparison new vs old model.

Repo Blueprint (Suggested Folder Structure)

root/	
■■■ api/	API endpoints (Next.js/Node/Python workers)
■■■ agents/	Multi-agent definitions & orchestration logic
■■■ data/	Eval sets, golden examples, adversarial prompts
■■■ dashboards/	Grafana/PostHog configs + screenshots
■■■ retraining/	Drift detection + retraining scripts
■■■ router/	Model routing logic + config policies
■■■ tests/	Unit tests, integration tests, red-team cases
■■■ README.md	Project overview + setup instructions